PriceHawk — Price Scraper + Alert Tool

Flagship project · Python · 20 automated tests · MIT license

What it does

- Scrapes any product listing page (multi-page, follows pagination)
- Exports clean CSV / JSON / styled Excel with one command
- Tracks every run as a timestamped price snapshot
- Diffs two snapshots: price drops, new items, removals — with % change
- Exit code 2 when prices change → drop into cron/CI for automatic alerts

Live demo — real output (books.toscrape.com, 2 pages)

```
$ pricehawk scrape http://books.toscrape.com/ \
      -o products.csv --history history.json --pages 2
  [page 1]  20 items   [page 2]  20 items
[stats] 40 items · price min 12.84 · max 57.25 · avg 34.96
[ok] exported 40 rows → products.csv

$ pricehawk diff --history history.json
PRICE CHANGES   53.74 → 43.89 (-18.3%)  Tipping the Velvet
NEW ITEMS       + Scouting for Wilderness Adventure
REMOVED         - The Coming Woman: A ...
```

Architecture

- cli.py — argparse commands (scrape / diff), proper exit codes
- scraper.py — fetch with retry + backoff · pure HTML parsers · pagination
- export.py — CSV / JSON / XLSX writers (frozen header, bold columns)
- history.py — snapshots + diff math + human-readable report
- Selectors live in JSON config → adapt to a new shop without code changes

Quality

- 20 unit tests, all green — price parsing (US$ / R$ / £), parser, pagination,
- export round-trip, snapshot history, diff math
- Polite scraping: custom UA, delay between pages, retries with backoff, timeout

Real screenshots — run & diff

Not mockups: captured from the actual CLI

pricehawk — scrape (live run)

```
$ pricehawk scrape http://books.toscrape.com/ \
    -o products.csv --history history.json --pages 2
PriceHawk 1.0.0 — scraping http://books.toscrape.com/
  [page 1]  20 items  http://books.toscrape.com/
  [page 2]  20 items  http://books.toscrape.com/catalogue/page-2.html
[stats] 40 items · price min 12.84 · max 57.25 · avg 34.96
[ok] exported 40 rows → products.csv
[ok] snapshot saved → history.json (2026-09-25T17:14:19.133472+00:00)
```

pricehawk — diff (price alerts)

```
$ pricehawk diff --history history.json
PRICE CHANGES
  37.59 → 30.53 (-18.8%)  Mesaerion: The Best Science ...
  53.74 → 43.89 (-18.3%)  Tipping the Velvet
NEW ITEMS
  + Scouting for Wilderness Adventure (Collins Little Books)
  + The Wilderness War (The Last Wild)
REMOVED
  - The Coming Woman: A ...
```

Real data output & how delivery works

40 books scraped live → table below is the actual CSV

Sample of the real exported data

REAL SCRAPE OUTPUT

40 books · live demo run

#	name	price	rating	stock
1	A Light in the ...	51.77	3/5	yes
2	Tipping the Velvet	53.74	1/5	yes
3	Soumission	50.10	1/5	yes
4	Sharp Objects	47.82	4/5	yes
5	Sapiens: A Brief History ...	54.23	5/5	yes
6	The Requiem Red	22.65	1/5	yes
7	The Dirty Little Secrets ...	33.34	4/5	yes
8	The Coming Woman: A ...	17.93	3/5	yes

books.toscrape.com · 2 pages · CSV + XLSX + history

Use cases

- Competitor price monitoring → scheduled CSV + price-drop alerts
- Catalog export / migration (any e-commerce platform)
- Lead lists, listings, jobs, reviews from public directories

Packages

- BASIC $30 · 1 page, up to 100 rows, 1 format · 3 days
- STANDARD $70 · up to 10 pages / 1,000 rows, CSV + XLSX + script · 5 days
- PREMIUM $150 · larger jobs, scheduled scrape + price history · 7 days

Rules I follow

- Public data only · respect robots.txt and Terms of Service
- No login bypass, no private/personal data harvesting
- Rate-limited requests — your target site stays healthy